

Senior Guide: Ollama Enterprise Implementation

Senior Implementation Guide: Ollama Production Bundle

This project requires a robust, local-first IA orchestration. You are responsible for the seamless integration of Ollama into the Electron environment.

Technical Requirements (Must-Have)

- Use `child_process.spawn` to manage "ollama serve" as a background worker.
- Implementation of a "Silent Installer Check": if ollama is not in PATH, run `resources/ollama/OllamaSetup.exe` in silent mode (/S or similar).
- Zero Latency Strategy: Use `OLLAMA_KEEP_ALIVE=-1` env variable when starting the service.
- API Standardization: Map all AI calls to the OpenAI-compatible endpoint (`/v1/chat/completions`).

Model Lifecycle Management

- Proactive Pulling: Check for `gemma2:2b` and `llama3.2-vision` availability via `/api/tags`.
- Non-blocking Download: Trigger "ollama pull" asynchronously and report progress via IPC to the frontend progress bar.

Error Resilience

- Health check: Ping `127.0.0.1:11434` before any inference.
- Graceful Fallback: If local GPU fails, switch automatically to Gemini API based on user settings.